

Goji berry effects on macular characteristics and plasma antioxidant levels.



PURPOSE: Goji berry (*Lycium barbarum* L.) is purported to benefit vision because of its high antioxidant (especially zeaxanthin) content, although this effect has not been demonstrated in high-quality human studies. The purpose of this study was to evaluate the effects of daily supplementation with a proprietary milk-based formulation of goji berry, Lacto-Wolfberry (LWB), on macular characteristics and plasma zeaxanthin and antioxidant capacity levels in elderly subjects. **METHODS:** This was a double-masked, randomized, placebo-controlled trial in healthy elderly subjects (range, 65 to 70 years) receiving 13.7 g/d of LWB (n = 75) or placebo (n = 75) for 90 days. Subjects underwent direct ophthalmic examination to assess pigmentation and soft drusen count in the macula and a blood draw to measure plasma zeaxanthin level and total antioxidant capacity. **RESULTS:** The placebo group demonstrated hypopigmentation and soft drusen accumulation in the macula, whereas the LWB group remained stable. Both plasma zeaxanthin level and antioxidant capacity increased significantly in the LWB group, by 26% and 57%, respectively, but did not change in the placebo group. No product-related adverse events were reported in either group. **CONCLUSIONS:** Overall, daily dietary supplementation with goji berry for 90 days increases plasma zeaxanthin and antioxidant levels as well as protects from hypopigmentation and soft drusen accumulation in the macula of elderly subjects. However, the mechanism of action is unclear, given the lack of relationship between change in plasma zeaxanthin and change in macular characteristics.